

RPAT

Ernest Sebastian - Section 6 - Density - Altitude - Assignment.

1. Calculate the density altitude of location.

Location details.

i) Airport

: Jomo Kenyatta Int. Airport (KE)

ii) Time of recording data

: 08:00 UTC

iii) The elevation of airport

: 1624m

iv) The temperature

: 19°C

v) The pressure

: 1026 hPa

∴ First we calculate the pressure altitude

Using 1026 hPa, altitude correction is -348 ft / 106.0664m

This means pressure altitude = 1624m - 106.0664m
= 1517.9m or 4977.6ft.

∴ Convert the elevation to feet and get ISA temp
1624m = 5327ft

∴ ISA Temp of 5327ft = 5.1°C

3. Calculate density altitude equation

$$\begin{aligned} P_{\text{density alt}} &= P_{\text{pressure alt}} + (120 * (T_{\text{ambient air temp}} - T_{\text{ISA at alt}})) \\ &= 4977.6 + (120 * (19 - 5.1)) \\ &= \underline{\underline{5805.6ft}} \end{aligned}$$

2 Effect that the altitude density may have on the aircraft's flight.

At 5806.6 ft, this clearly means that the air at Jomo Kenyatta Int. Airport in Kenya is less dense. Such being the condition, the aircraft is impacted in several ways during takeoff, landing, engine performance and more.

On takeoff performance, the high altitude means the engine power is reduced thereby lift is also decreased and with that, the time it takes to takeoff is increased and it takes a very long time for the aircraft to takeoff on the runway.

During landing the aircraft can require a longer runway to match up the lift that is decreased due to the reduced air density. The engine may also need to work more as its performance is reduced due to the lower air density. This greatly impacts the aircraft performance.